INCH-POUND

A-A-52300
March 3, 1992

COMMERCIAL ITEM DESCRIPTION

PROTRACTORS, CIRCULAR AND SEMICIRCULAR, (MILS, DEGREES)

The General Services Administration has authorized the use of this commercial item description in preference to Federal specification GG-P-681D.

This description covers circular protractors of plastic and semicircular protractors of brass or plastic for use in graphically measuring precise angles required by mapping and precision drafting techniques.

Salient characteristics:

Design. The protractors shall be as specified herein. The protractors shall have precise graduations, letters, numbers, and shall be constructed of materials (brass or plastic) designed to lie flat on a plane surface without stretching or warping. All graduation, letters, and numbers shall be complete, sharp, and precisely formed.

Classification.

Types, classes, styles, and sizes. Protractors shall be of the following types, classes, styles, and sizes.

Type I - Semicircular.
Class A - Graduated in degrees.
Class B - Graduated in mils.
Style 1 - Without arc extension.
Style 2 - With arc extension.

Sizes (diameter).
4 inches.
6 inches.
8 inches.
10 inches.
12 inches.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent by letter to: Commander, U.S. Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 6675

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Type II - Circular, graduated in degrees.
Style 1 - Zero to 360 degrees.
Style 2 - Zero to 90 degree quadrants and zero to 360 degrees.

Sizes (diameter).
6 inches.
8 inches.
10 inches.

Type I, semicircular. Type I protractors shall be of brass or plastic. Numbers may be slanted in the direction of increase. The outer row of numbers on the graduated arc shall read clockwise or counterclockwise and the inner row of numbers shall read in the opposite direction.

Class A. The outer edge of the arc shall be graduated in degrees so that the zero and 180 degree marks are on the horizontal axis of the protractor. When the upper edge of the crossbar forms the horizontal axis, the graduations may be extended below that axis. The graduation marks that extend below the zero and 180 degrees marks of the horizontal axis are the arc extension.

Class B. The outer edge of the arc shall be graduated in mils so that the zero and 3200-mil marks are on the horizontal axis of the protractor. When the upper edge of the crossbar forms the horizontal axis, the graduations may be extended below that axis.

NOTE: A mil is defined as 1/6400 of a circle. The graduation marks for protractors reading mils shall be not less than three lengths; the shortest length denoting 10-mils, the medium length denoting each 50-mil increment, and the longest denoting each 100-mil increment. Graduation marks shall be numbered every 100-mils in two rows of figures.

Style 1. Configuration of style 1 protractors shall be similar to that shown on figure 1.

Style 2. Configuration of style 2 protractors shall be similar to that shown on figure 2.

Type II, circular. Type II protractors shall be of plastic. Numbers may be slanted in the direction of increase.

Style 1. The outer row of figures shall read either clockwise or counterclockwise from zero to 360 degrees, starting at the horizontal axis. The inner row of figures shall read in the direction opposite that of the outer row, starting on the opposite end of the horizontal axis. Configuration of type II, style 1 protractors shall be similar to figure 3.

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Style 2. The outer row of figures shall read clockwise from zero to 90 degrees; 90 to zero degrees; zero to 90 degrees; and 90 to zero degrees, starting at the left in line with the horizontal axis. The inner row of figures shall read clockwise from zero to 360 degrees, starting at the left in line with the horizontal axis.

Materials:
Material shall be as specified herein.

Brass. Brass shall be a flat rolled sheet up to and including 0.188-inch in thickness. Brass sheet tempered to half-hard shall be acceptable.

Plastic. Plastic protractors shall be made out of acrylic, methacrylate, or any other plastic material that provides similar durability.

Graduations, letters, and numbers. All graduations, letters, and numbers shall be cut, stamped, printed, photoprinted, etched, or indented. Graduation lines shall be uniform, shall be between 0.004 and 0.010-inch wide, and shall have a tolerance of ±5 minutes of arc. Graduations, letters, and numbers shall not project above the surface of the brass or plastic. Unless otherwise specified, cut, stamped, etched, or indented graduations, letters, and numbers shall be filled with permanent black ink or paint.

Degrees. The graduation marks for protractors reading in degrees shall be of not less than four lengths; the shortest length denoting 0.5 and the longest denoting each 10-degree increment. Graduation marks shall be numbered every 10 degrees in two rows of figures. The 0.5-degree graduation is not required on 4-inch protractors.

Crossbar. Each protractor shall have a crossbar formed by removing a semicircular segment or segments from the protractor so that the width of the crossbar and the section between the concentric arcs are equal. A vertical line shall be formed on the crossbar, intersecting the horizontal axis, to indicate the vertical axis of the protractor.

Brass protractors. Brass protractors shall be flat and shall be between 0.03125 and 0.0625-inch thick except that 4-inch protractors shall be more than 0.015625-inch, but less than 0.0625-inch thick. The horizontal axis shall be along either edge of the crossbar. A vertical line shall be formed on the crossbar to indicate the vertical axis of the protractor. The line shall be formed by the same method as that used for the graduations.

Plastic protractors. Plastic protractors shall be flat and shall be between 0.0625 and 0.125-inch thick. Graduations shall be on the bottom surface of the plastic and shall be readily distinguishable through the thickness of plastic. Letters and numbers may be on either surface of

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the protractor but shall be in reading position when the graduated surface is placed in contact with the drawing, map, or other work being performed.

Horizontal axis. On plastic protractors, the horizontal axis may be along either edge of the crossbar or within the area between the edges of the crossbar. When the horizontal axis is formed by either edge of the crossbar, a bevel not less than 0.500-inch long shall be formed on the top surface of the crossbar along the horizontal axis, centered with the line indicating the vertical axis of the protractor. The lower edge of the bevel shall be not less than 0.015-inch nor more than 0.031-inch from the bottom surface of the protractor at the point of intersection of the vertical and horizontal axis. When the horizontal axis falls within the area between the edges of the crossbar, each axis shall be represented with a line formed by the same method as that used for the graduations, and at the point of intersection of the horizontal and vertical axis a 0.0625-inch diameter hole shall be drilled, countersunk or formed. The outer edge of the protractor shall be beveled on the top surface of the graduated edge.

Workmanship. The surface and edges of finished protractors shall be true, smooth, uniform, and polished and shall be free from defects which affect the appearance or which might impair the serviceability or use of the protractor.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Commercial Item Description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Quality assurance. The circular and semicircular protractors shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

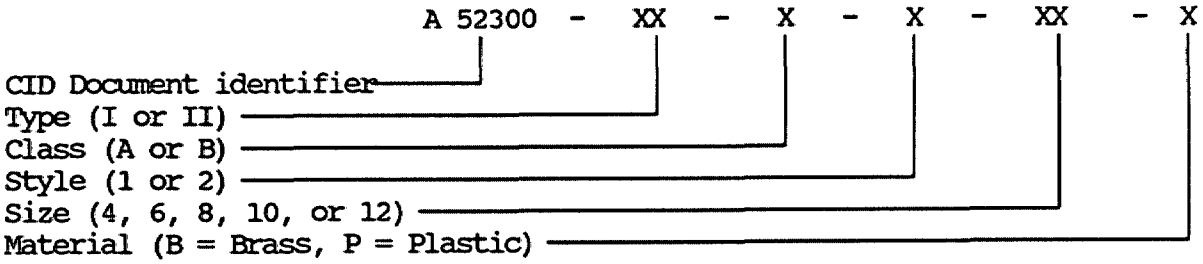
1. Design not as specified.
2. Type not as specified.
3. Class not as specified.
4. Style not as specified.
5. Size not as specified.
6. Graduations, letters, and numbers not as specified.
7. Degrees not as specified.
8. Crossbar not as specifieded.
9. Brass protractors not as specified.
10. Plastic protractors not as specified.
11. Horizontal axis not as specified.
12. Material not as specified.
13. Workmanship not as specified.
14. Packaging and marking not as specified.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Part or identifying number (PIN). The part or identifying number for protractors, circular and semicircular, (mils, degrees) shall be as follows:



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Notes. The procuring agency should specify the preferred options permitted herein and include the following information in procurement documents:

1. Title, number, and date of this item description.
2. When the commercial item description is issued for acquisition, the commercial item certification clause must appear in the solicitation.
3. The ordering activity shall specify type, class, style, size, and material required.

FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

The following is a suggested source of supply for this item description (Competition is not limited to this source.):

1. Lutz Superdyne, Inc., 815 Broad Ave., P.O. Box 57, Ridgefield, NJ 07657.

The preparing activity for this item description is the US Army Belvoir Research Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

MILITARY INTEREST:

Custodians
Army - ME
Air Force - 99

Review activity
DLA - GS
Navy - MC
Air Force - 82

User activities
Navy - OG

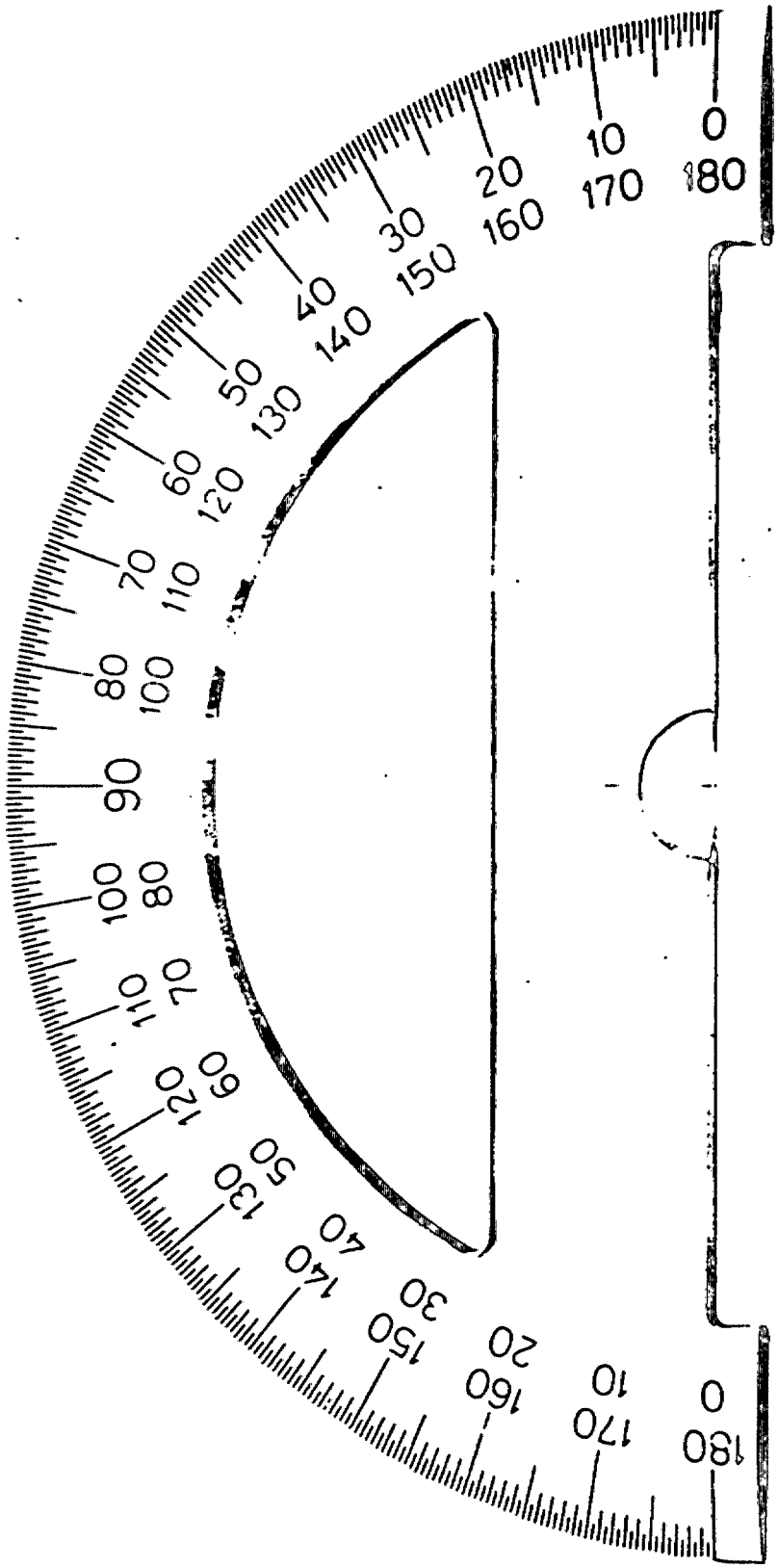
CIVIL AGENCY COORDINATING ACTIVITIES.

GSA-FSS

PREPARING ACTIVITY:
Army - ME

Project 6675-0895

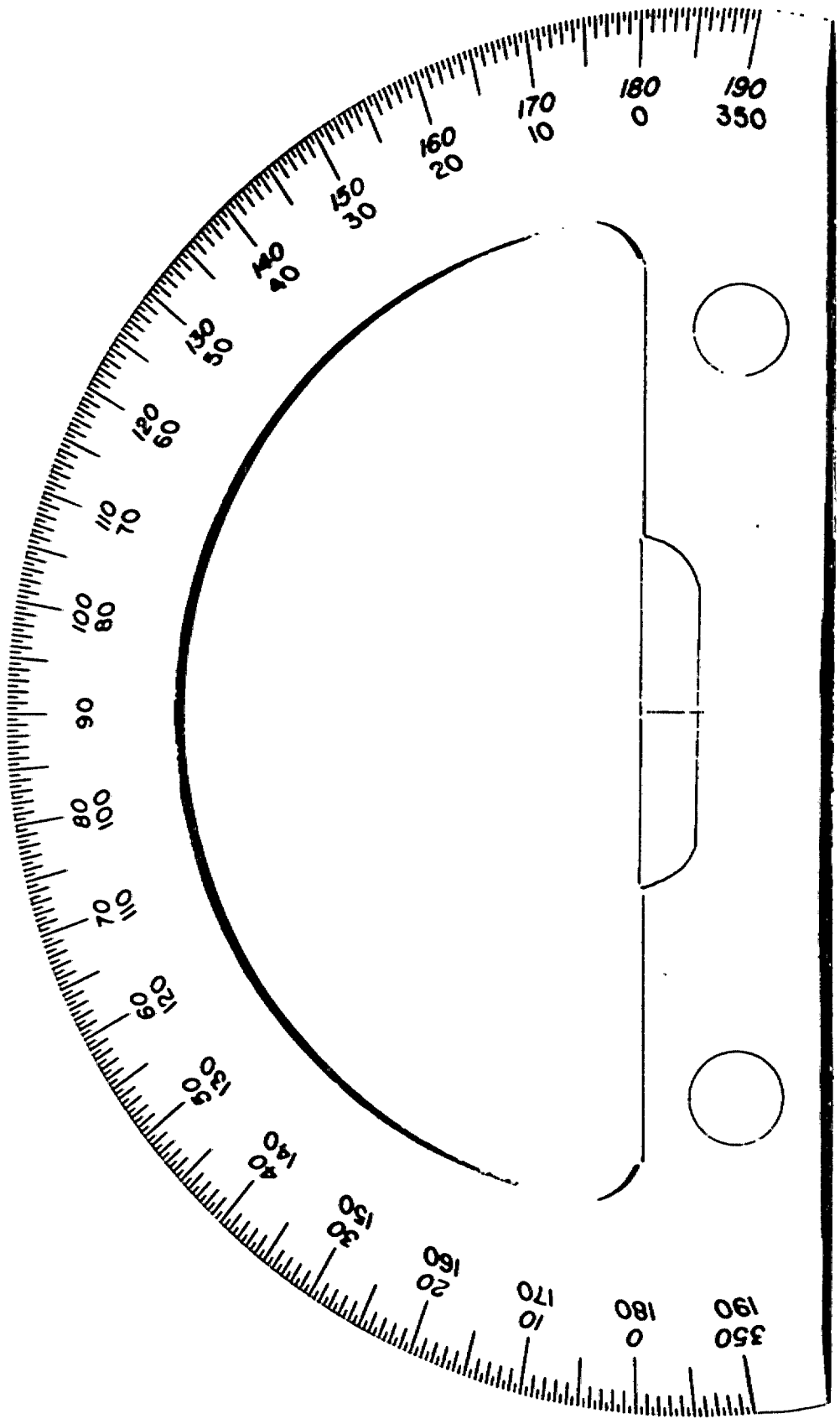
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FIGURE 1. Type I, class A, style I protractor without arc extension.

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FIGURE 2. Type I, class A, style 2 protractor with arc extension.

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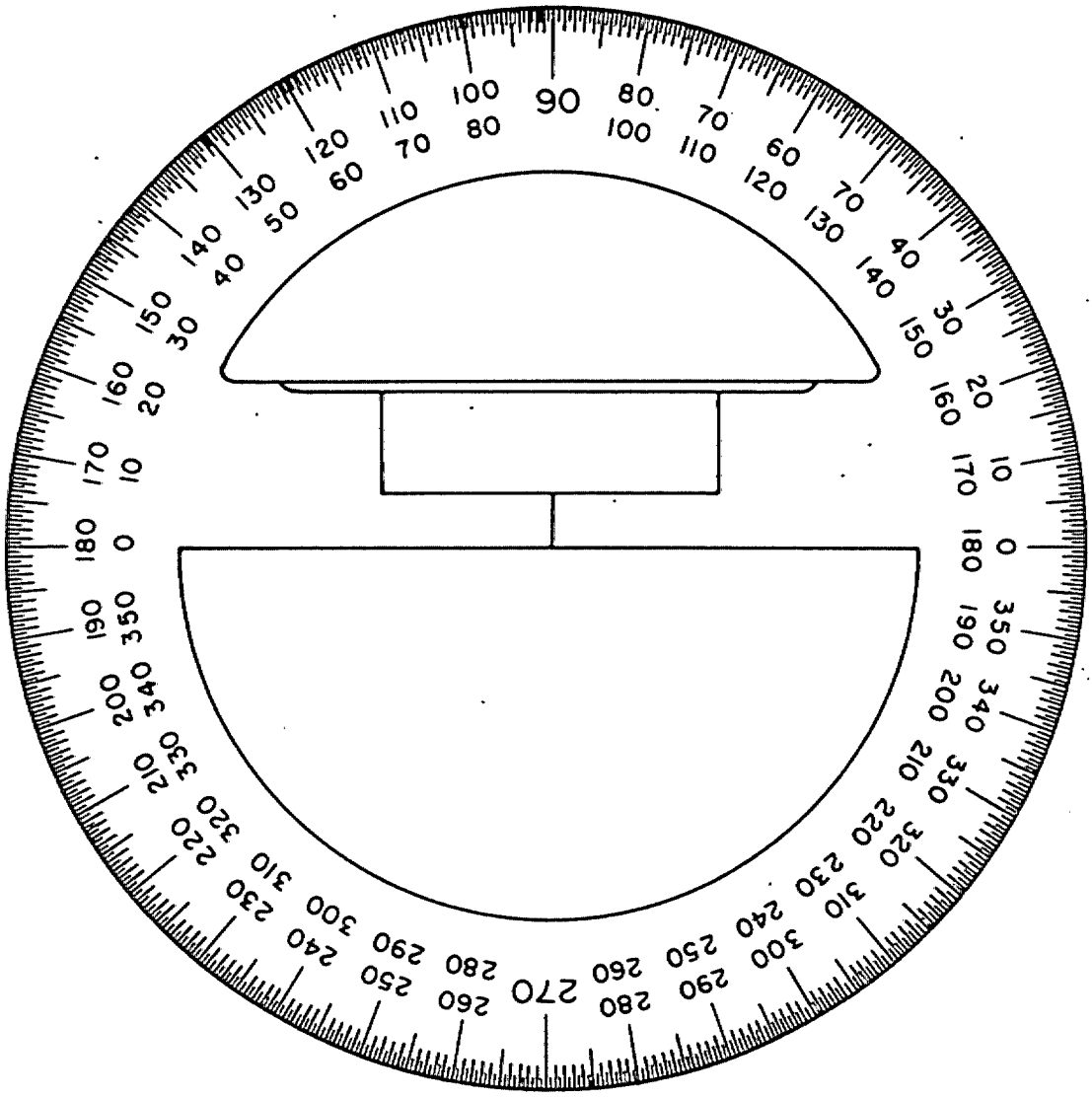


FIGURE 3. Type II, style I, protractor.

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